

1. RUN CONFIGURATION

Frames Processed	Duration	Processing FPS	Avg Processing	Max Processing
6	0.05 s	132.3	2.00 ms	2.97 ms

2. TRACKING PERFORMANCE

Acquisition Time	Average Tracking Error	Maximum Tracking Error
0.03 s	3.16 px	3.16 px
<= 2 s		<= 10 px
RMSE Tracking Error	Centroid Error Avg/Max	Lock Retention
3.16 px	3.16 / 3.16 px	— %
	ground truth only	
Target Loss	Re-acquisition Avg/Max	Successful Re-acq.
— %	—	0
< 5 %	<= 1 s	0 loss events

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Acquisition Time	0.03	<= 2.0 s	PASS
Tracking Error (Max)	3.16	<= 10 px	PASS
Center Error (Final)	204.68	<= 10 px	FAIL
Target Loss	—	< 5%	PENDING
Re-acquisition Time	—	<= 1.0 s	PENDING
Processing Speed	132.3 FPS	>= 20 FPS	PASS

Ground truth was available: centroid accuracy is measured.

OVERALL: FAIL

Source: benchmark video: synthetic_fsoc_beacon.mp4 | Frames: 6 | FPS: 132.3